

Convergent retention of oxidative-defense orthogroups across independent extremotolerant origins: A comparative-proteomics screen in Deinococcaceae, Tardigrada, and Bdelloidea

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1 Introduction (skeleton)

Extremotolerant organisms have evolved radiation resistance independently at least three times: in the bacterial family Deinococcaceae, in tardigrades, and in bdelloid rotifers. Whether these origins converged on the same molecular machinery is unknown. We test whether orthogroups are convergently retained across resistant lineages while absent from radiation-sensitive controls.

2 Background: The Biology of Extreme Radiation Resistance

2.1 Ionizing radiation as a selective force

Ionizing radiation damages cells through two routes. Direct deposition of energy in DNA produces double-strand breaks (DSBs), the most lethal common lesion; a single unrepaired DSB can be sufficient to kill a dividing cell. Indirect damage is mediated by water radiolysis, which generates hydroxyl radicals, superoxide, and hydrogen peroxide, and these reactive oxygen species (ROS) oxidize proteins, lipids, and nucleic acids far from the site of the original photon absorption. For desiccation-tolerant organisms the indirect route dominates: at low water activity the primary lesion burden shifts toward oxidative protein carbonylation, and survival correlates more strongly with proteome protection than with DNA repair capacity alone.

2.2 The canonical model: *Deinococcus radiodurans*

D. radiodurans R1 survives acute gamma doses above 10 kGy without loss of viability, roughly three orders of magnitude beyond the lethal dose for *Escherichia coli*. Three decades of work have established that its resistance is not attributable to unusual target hardening of DNA. Instead, a layered defense acts on the proteome and the metabolome: (i) manganese accumulation, with intracellular Mn(II) concentrations reaching millimolar levels and Mn/Fe ratios far above those of sensitive bacteria; (ii) Mn(II) complexes with orthophosphate and small metabolites (peptides, nucleosides) that scavenge superoxide and hydroxyl radicals non-enzymatically; (iii) an unusually deep enzymatic antioxidant battery, including multiple catalases, superoxide

dismutases, and peroxiredoxins of the AhpC/TSA family; and (iv) an efficient homologous-recombination machinery operating on a multipartite, polyploid genome that reassembles shattered chromosomes from overlapping fragments through extended synthesis-dependent strand annealing. The resulting picture is that of a cell whose DNA is damaged at the same rate as any other cell's but whose proteins remain functional throughout irradiation, permitting rapid repair once conditions allow.

2.3 Convergent resistance in other clades

Extreme resistance has evolved independently in several lineages, and each case is an independent natural experiment. Within bacteria, the Deinococcaceae contain multiple resistant species (*D. radiodurans*, *D. geothermalis*, *D. deserti*), and the family is treated in this study as a single origin because of shared ancestry. Among animals, tardigrades such as *Ramazzottius varieornatus* tolerate kilogray doses in the desiccated state, and at least one lineage-specific protein, Dsup, binds nucleosomes and reduces DNA damage in heterologous systems, including human cells. Bdelloid rotifers such as *Adineta vaga* combine desiccation tolerance with radiation resistance and carry heavily reorganized genomes with signatures of horizontal gene transfer, including genes of bacterial and fungal origin implicated in stress response. These three origins - Deinococcaceae, Tardigrada, and Bdelloidea - are separated by more than a billion years of evolution, so any molecular feature shared across them is a candidate convergence on a common solution rather than a shared inheritance.

2.4 Why oxidative defense is the leading axis

Across resistant taxa the recurring observation is elevated antioxidant capacity: peroxiredoxins, glutaredoxin and thioredoxin systems, flavin reductases that recycle redox cofactors, and metal-homeostasis systems that limit iron-catalyzed Fenton chemistry. Peroxiredoxins of the AhpC/TSA family are peroxide-scavenging enzymes with exceptionally high catalytic rates; their up-regulation is a common response to irradiation in bacteria, and their abundance correlates with desiccation tolerance in tardigrades and rotifers. If radiation resistance across kingdoms shares a single dominant mechanistic axis, comparative genomics should recover orthogroups in this pathway as convergently retained or expanded in resistant lineages while absent from sensitive relatives - the hypothesis tested here.

2.5 The specificity problem

A central confound is that desiccation, oxidation, heat, and radiation stresses are correlated in nature. Organisms that survive drying necessarily solve the oxidative-damage problem, and radiation resistance may in part be a byproduct of desiccation adaptation. Any genomic enrichment found in resistant species could therefore reflect general stress adaptation rather than radiation-specific selection. This study addresses the confound directly with a pre-registered, stress-matched control panel (Section ??): stress-tolerant but radiation-sensitive species are subjected to the identical pipeline, and candidate orthogroups must be absent from that panel to retain a radiation-specific interpretation.

2.6 Known mechanisms of ionizing-radiation resistance

Decades of work on *Deinococcus radiodurans* established that extreme resistance is not conferred by a single pathway. Three mechanistic layers are recognized. First, an exceptionally clean antioxidant defense: high cellular $\text{Mn}^{2+}/\text{Fe}^{2+}$ ratios, manganese-phosphate metabolite complexes, and small-molecule antioxidants protect proteins — and especially the DNA-repair enzymes themselves — from oxidative inactivation, so that an otherwise conventional repair toolkit keeps working at doses that disable it in sensitive cells. Second, efficient and redundant DNA-repair capacity: extended synthesis-dependent strand annealing, RecA-dependent homologous recombination, and non-homologous end-joining reassemble chromosomes shattered into hundreds of fragments. Third, genome organization and proteome protection: nucleoid condensation, redundant genome copies, and chaperone/protease systems limit and clear damaged protein. Tardigrades add distinct intrinsically disordered protective proteins (CAHS, SAHS, and the DNA-shielding Dsup), demonstrating that more than one molecular route to resistance exists; bdelloid rotifers combine desiccation-linked antioxidant defense with horizontal-gene-transfer-enriched damage-response repertoires. A comparative screen across independent origins therefore asks not whether a single known factor is shared — none is — but which *modules* were independently retained, which is the question the locked gates operationalize.

2.7 Why convergence is the informative signal

Single-organism studies cannot separate general stress biology from radiation-specific adaptation, and single-lineage comparative genomics cannot separate ancestry from adaptation. Convergent retention across independent extremotolerant origins — here a bacterial family and a metazoan class — restricts attention to modules that selection kept rebuilding or re-keeping under the same pressure, while the stress-matched radiation-sensitive control panel removes modules that merely mark hardy organisms in general. This design logic follows the comparative strategy that identified convergent desiccation machinery, applied here to the radiation axis with pre-registered gates so that every retained or discarded candidate carries a stated reason.

3 Methods (pre-registered; amendments dated 2026-09-26)

Panel: 6 resistant proteomes (*Deinococcus radiodurans*, *D. geothermalis*, *D. deserti* [origin 1: *Deinococcaceae*]; *Ramazzottius varieornatus* [origin 2: *Tardigrada*]; *Adineta vaga* [origin 3: *Bdelloidea*]) vs 6 sensitive controls (*Thermus aquaticus*, *T. thermophilus*, *Hypsibius exemplaris*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Escherichia coli*). *Milnesium tardigradum* has no reference assembly; *Tardigrada* origin is single-proteome (power caveat carried throughout).

Orthogroup calling: MMseqs2 v17 sensitive clustering (min_seq_id 0.3, coverage 0.5, cov-mode 1), 154,665 proteins. Linclust used only as sensitivity; subset concordance (20k proteins): linclust pair purity 0.914 vs sensitive, retention 0.363, so all formal claims rest on sensitive clustering.

Formal test (locked 11:15 and 11:29 amendments, before outcome inspection): per orthogroup, one-sided Fisher exact on presence/absence (6 resistant vs 6 sensitive) and one-sided Mann-Whitney U on zero-filled copy numbers. The BH-FDR gate was replaced pre-outcome by 1000 label permutations with per-orthogroup empirical p , after W03's power analysis showed BH-FDR is structurally impossible at 6-vs-6 (minimum achievable Fisher

$p = 1/\binom{12}{6} = 1.08 \times 10^{-3}$ vs BH rank-1 threshold $\approx 10^{-6}$). Gate: empirical $p < 0.05$, resistant-direction, replication in ≥ 2 of 3 independent resistant origins.

Candidate annotation: NCBI esummary + hmmscan (HMMER 3.3.2) vs Pfam-A (full-sequence $E < 10^{-10}$).

3.1 Formal framework

Each orthogroup g is a partition of the panel proteomes into member sets; for species i let $n_{gi} \geq 0$ be the copy number and $x_{gi} = \mathbf{1}[n_{gi} > 0]$ the presence indicator. The resistant and sensitive label vectors $y \in \{0, 1\}^{12}$ are fixed by the panel definition. The presence/absence arm applies a one-sided Fisher exact test to the 2×2 table of x_g against y ; under complete separation (all resistant present, all sensitive absent) the minimum achievable value is

$$p_{\min} = \frac{1}{\binom{12}{6}} = 1.08 \times 10^{-3}, \quad (1)$$

which motivated replacing the BH-FDR gate by permutation calibration (pre-outcome amendment). The empirical p -value for orthogroup g over $B = 1000$ label permutations π_b is

$$\hat{p}_g = \frac{1 + \sum_{b=1}^B \mathbf{1}[T(x_g, \pi_b(y)) \geq T(x_g, y)]}{1 + B}, \quad (2)$$

with statistic T the resistant-direction enrichment score; the floor resolution for complete-separation configurations is $\hat{p} = 1/(1 + B) \approx 10^{-3}$ granularity at the observed $\hat{p} = 0.023$. The copy-number arm uses a one-sided Mann-Whitney statistic

$$U_g = \sum_{i: y_i=1} \sum_{j: y_j=0} \mathbf{1}[n_{gi} > n_{gj}] + \frac{1}{2} \mathbf{1}[n_{gi} = n_{gj}] \quad (3)$$

on zero-filled copy numbers. The convergence gate additionally requires replication across independent origins:

$$\sum_{k \in \{\text{Deino, Tardi, Bdello}\}} \mathbf{1} \left[\max_{i \in k} x_{gi} = 1 \right] \geq 2. \quad (4)$$

For the specificity layer, each v1 candidate was re-clustered with four stress-tolerant but radiation-sensitive controls and re-tested identically; a candidate keeps the label radiation-specific only if

$$|\{c \in \text{stress controls} : x_{gc} = 1\}| \leq 1, \quad (5)$$

otherwise it is re-labeled broad-stress adaptation. Domain-level confirmation searches each candidate's full Pfam architecture $\mathcal{D}_g = \{d_1, \dots, d_m\}$ against the five confirmation proteomes with family HMMs, accepting a proteome as supporting iff

$$\exists \text{protein } q : \bigwedge_{d \in \mathcal{D}_g} [E(q, d) \leq 10^{-5}], \quad (6)$$

and confirming the candidate iff at least two of five proteomes support it. The phylogenetic layer (H4–H4c) maps presence onto a fixed cladogram τ and counts minimum gains by Fitch reconstruction,

$$N_{\text{ind}}(g) = \min_{\text{ASR}} \sum_{e \in \tau} \mathbf{1}[0 \rightarrow 1 \text{ on } e, e \text{ subtends a resistant leaf}], \quad (7)$$

against a null of 10^4 within-strata presence shuffles σ_r preserving stratum counts,

$$p_{\text{lower}}(g) = \frac{1 + \sum_r \mathbf{1}[N_{\text{ind}}^{\sigma_r} \leq N_{\text{ind}}(g)]}{1 + 10^4}, \quad (8)$$

which proved degenerate on saturated strata (zero null variance), yielding the identifiability boundary reported in Section 3.4.

4 Results

Of 52,786 orthogroups tested, **12 pass all locked gates**: convergently present in Deinococcaceae and Bdelloidea, absent from all 6 sensitive controls, empirical $p = 0.023$ (1000 permutations). A further 53 orthogroups pass the copy-number Mann-Whitney arm.

Table 1: Convergent orthogroups passing all locked gates, with Pfam domain calls.

Representative	Pfam domains (E-value)	Interpretation
A. vaga UJR09360.1	AhpC-TSA (3.8e-35), Redoxin (8.8e-17)	Peroxioredoxin; peroxide detox
D. geothermalis WP_272975990.1	DSBA (7e-50)	Thiol-disulfide oxidoreductase
D. deserti WP_012692142.1	Flavin_Reduct (8.1e-26)	Flavin reductase; redox
D. deserti WP_012692117.1	Ank $\times 5$ (6e-24...)	Ankyrin scaffold; DDR interactions
D. geothermalis WP_414646549.1	PAD_porph (7.7e-122)	Porphyrin/heme binding
D. geothermalis WP_272975981.1	Acetyltransf_1/7	GNAT acetyltransferase
A. vaga UJR10666.1	HPPK (3.3e-29), Pterin_bind	Folate/pterin pathway
A. vaga UJR11615.1	DHH (2.8e-21), DHHA2 (2e-22)	Phosphoesterase; nucleotide damage
A. vaga UJR26281.1	HAD_2 (2.2e-25), Hydrolase	HAD-family hydrolase
A. vaga UJR28130.1	GDP_Man_Dehyd, NmrA	Redox-sensing dehydrogenase
A. vaga UJR36058.1	DUF4385 (2.6e-60)	Uncharacterized (novel space)
A. vaga UJR17759.1	Amidase (2.8e-61), Bac_rhamnosid6H	Cell-wall/envelope metabolism

The annotated candidates concentrate on the oxidative-defense axis: peroxiredoxin (AhpC-TSA), thioredoxin-family oxidoreductase (DSBA), and flavin reductase. Ionizing-radiation damage is dominated by reactive oxygen species; convergent retention of an oxidative-defense module across a bacterial and a metazoan origin, with complete absence from sensitive controls, is the screen’s central signal. The folate/pterin candidate (HPPK) is consistent with the known *Deinococcus* folate-photoprotection mechanism, and DHH/DHHA2 phosphoesterases act in nucleotide-damage processing.

4.1 Specificity layer on the extended panel (amendment 2026-09-26, 13:18 IST)

The panel was extended pre-data with four stress-tolerant but radiation-sensitive controls — *Pyrococcus furiosus*, *Sulfolobus solfataricus*, *Lactobacillus plantarum*, and *Saccharomyces cerevisiae* — chosen to separate radiation-specific retention from generic stress adaptation. All 168,186 proteins from the 15 proteomes were re-clustered de novo at the locked parameters (min. seq. id. 0.3, coverage 0.5, cov-mode 1), yielding 60,164 clusters. The locked gates were re-run identically on this v2 clustering, and each v1 candidate was then required to be absent from at least 3 of the 4 new controls to retain the label *radiation-specific*; otherwise it is re-labeled *broad-stress adaptation*.

All 12 v1 representative proteins mapped into v2 clusters. **10 of 12 candidates pass both the re-run enrichment gate (empirical $p = 0.013$ – 0.017 , 1000 permutations) and the specificity layer**, each with Deinococcaceae+Bdelloidea origins and at most one control-species member. Two honest negatives: (i) the v1 lead, the AhpC-TSA peroxiredoxin (UJR09360.1), is present in 3 of 4 stress-tolerant controls and is re-labeled broad-stress adaptation — biologically consistent, since peroxiredoxins are near-universal stress proteins; (ii) the HPPK/pterin candidate (UJR10666.1) lost its Deinococcaceae members in the larger panel (Bdelloid-only cluster, empirical $p = 0.449$) and no longer shows cross-kingdom convergence.

The oxidative-defense axis survives the specificity layer through the DSBA thiol-disulfide oxidoreductase and the flavin reductase candidates, now supported as radiation-specific rather than generic stress adaptations. The surviving set additionally includes the ankyrin scaffold, PAD-porphyrin binding protein, GNAT acetyltransferase, DHH/DHHA2 phosphoesterase, HAD hydrolase, GDP-mannose dehydrogenase, amidase, and the uncharacterized DUF4385 protein — the last representing genuinely novel candidate space.

4.2 Domain-level orthogonal confirmation (amendment 2026-09-26, 20:21 IST)

As an independent check, the panel was extended with five radiation-resistant proteomes never used in the screen: *Deinococcus radiophilus*, *D. proteolyticus*, *D. maricopensis*, *Adineta ricciae*, and *Rotaria magnacalcarata* (95,758 proteins; accessions and hashes in the ledger). For each radiation-specific candidate, its complete v1 Pfam domain set was searched against all five new proteomes (64 family HMMs; domain calls at $E \leq 10^{-5}$, fixed before candidate-level results). **9 of 10 candidates are confirmed:** their full domain architectures recur in at least one protein of at least two of the five new proteomes (eight at 5/5 support, one at 4/5). The five-domain amidase architecture (UJR17759.1) found no complete match and is labeled “domain-level support absent in extended panel” without removal from the candidate set (locked rule); the HPPK candidate that failed v2 enrichment is likewise unconfirmed, a consistent picture.

4.3 Phylogenetically controlled convergence test (amendments H4–H4c, 2026-09-26)

The strongest remaining objection to the v2 labels is phylogenetic confounding: the three *Deinococcus* proteomes (like the two *Thermus* controls) are not independent replicates, so a label permutation that treats species as exchangeable only partially randomizes family-level structure. We therefore implemented a phylogenetically controlled convergence test, locked before computation (amendment H4): for each radiation-specific candidate, presence/absence was mapped onto a fixed 15-species cladogram (topology from NCBI taxonomy, unit branch lengths), and the minimum number of independent gains into resistant-subtending branches was computed by Fitch-parsimony ancestral-state reconstruction (the minimum-gain count is method-independent; a PastML maximum-likelihood cross-check is queued). The null was 10,000 structure-preserving permutations that shuffle presence assignments only within exchangeable strata (Deinococcaceae, *Thermus*, other controls), never across. Formally, let $x_g \in \{0, 1\}^{|\text{leaves}(\tau)|}$ be the presence vector of candidate g on the fixed cladogram τ . The Fitch bottom-up pass assigns each internal node v the intersection of its children’s state sets when non-empty, else their union, and the minimum-gain statistic is the number of union events,

$$G(x_g; \tau) = \sum_{v \in \text{int}(\tau)} \mathbf{1}[S_v = S_{v_1} \cup S_{v_2}], \quad (9)$$

where S_v is the Fitch state set of node v with children v_1, v_2 , and a union event implies exactly one gain on the branch subtending the child absent the state. The structure-preserving null draws permutations π_b that shuffle presence labels only within exchangeable strata $\mathcal{S} \in \{\text{Deinococcaceae}, \text{Thermus}, \text{other controls}\}$, and the corrected (lower-tail, amendment H4c) empirical convergence p -value is

$$\hat{p}_g^{\text{conv}} = \frac{1 + \sum_{b=1}^B \mathbf{1}[G(\pi_b(x_g); \tau) \leq G(x_g; \tau)]}{1 + B}, \quad (10)$$

with $B = 10,000$; the lower tail is the convergence-consistent direction because scattered presence forces *more* gains, so unusually *few* gains reject the structured null.

The test returned a structural null: for all ten candidates the null distribution has zero variance (observed 2 gains, null median 2, $p = 1.0$). The cause is identifiability, not biology: every candidate is present in all three Deinococcaceae, and the Tardigrada and Bdelloidea origins are singletons whose labels cannot be permuted, so no within-strata shuffle can move the signal. Resistance labels are perfectly confounded with tree topology on this panel.

We then locked an extended-panel arm (amendment H4b) adding the five confirmation lineages (three further Deinococcaceae, two further bdelloids; presence at the domain level only, per the confirmation-arm rule), and, after inspection revealed a tail-direction bug in our own locked statistic (scattered presence requires *more* gains, so the convergence-consistent direction is the lower tail), corrected it under a new locked amendment (H4c) before re-evaluation. On the corrected test, 9 of 10 candidates remain degenerate — saturated strata shuffle identically, so the null is constant — and the single candidate with partially occupied strata (the five-domain amidase UJR17759.1; 3/6 Deinococcaceae, 1/3 bdelloids) shows 2 observed gains against a null median of 3 ($p = 0.299$, not significant).

Identifiability boundary (quantified; redirect in progress). On a panel whose resistant lineages form saturated clades, the structured-presence-permutation test implemented here cannot separate convergence from clade structure; power exists only for phylogenetically dispersed or partially occupied strata. The strengthened claim “convergence beyond what species relatedness predicts” is therefore *not* earned by any candidate at this stage, and none receives it. All ten candidates retain their v2 radiation-specific labels with an explicit phylogenetic-structure caveat (locked rule: the confirmation layer never removes candidates). This boundary is not accepted as terminal: a redirection is in progress on three axes - (i) test designs that escape the saturation degeneracy on the same data (gene-tree/species-tree reconciliation, selection-regime comparisons on candidate orthogroups, phylogenetic regression of resistance on presence); (ii) additional phylogenetically dispersed resistant genomes that make the structured null non-degenerate; (iii) reframings in which clade-saturating cross-kingdom presence itself becomes the testable evidence. Whichever redesign is adopted will be locked as a dated amendment before evaluation. Until then, this quantified boundary stands as the primary limitation of the study, replacing our earlier qualitative admission (caveat 3).

5 Caveats (honesty section)

(1) Tardigrada origin is single-proteome (Milnesium terminally unavailable); the gate’s “ ≥ 2 origins” was met by Deinococcaceae+Bdelloidea in every passing orthogroup - no Tardigrada-supported convergence is claimed. (2) Empirical $p = 0.023$ is the floor resolution at 1000 permutations for complete-separation configurations; it is not a continuous measure. (3) Empirical- p calibration assumes exchangeability under label permutation; family-level phylogenetic structure is only partially randomized away. Section 3.4 quantifies this: structured-presence-permutation

tests are degenerate on saturated clades, so convergence beyond relatedness is not claimed for any candidate (terminal identifiability statement). (4) 4 esummary records failed uid resolution and await retry; their Pfam calls stand via hmmscan. (5) Candidates are computational; no experimental validation is claimed. (6) The v1 headline peroxiredoxin did not survive the specificity layer (Section 3.1); v1-era claims about AhpC-TSA as radiation-specific are superseded by the v2 labels.

6 Next experiments (falsifiable)

(1) Heterologous expression of the AhpC-TSA orthogroup in *E. coli*: gamma-survival assay. (2) Tardigrade proteome addition when a second assembly appears; re-run with identical locked gates. (3) Domain-level convergence test (hierarchy step 2 of the design amendment) as the orthogonal confirmation.